STAT 432 - Homework 02

Lu Gong

Question 1. Training and test error

(a) Simulation results

I used seed 43202 to generate one 100 by 20 normal design matrix and kept it fixed. For each of 200 runs, I generated independent training and test errors, fitted an intercept and the first *m* predictors, and divided each sum of squared errors by 100. The test response was evaluated at the same design rows, with new noise.

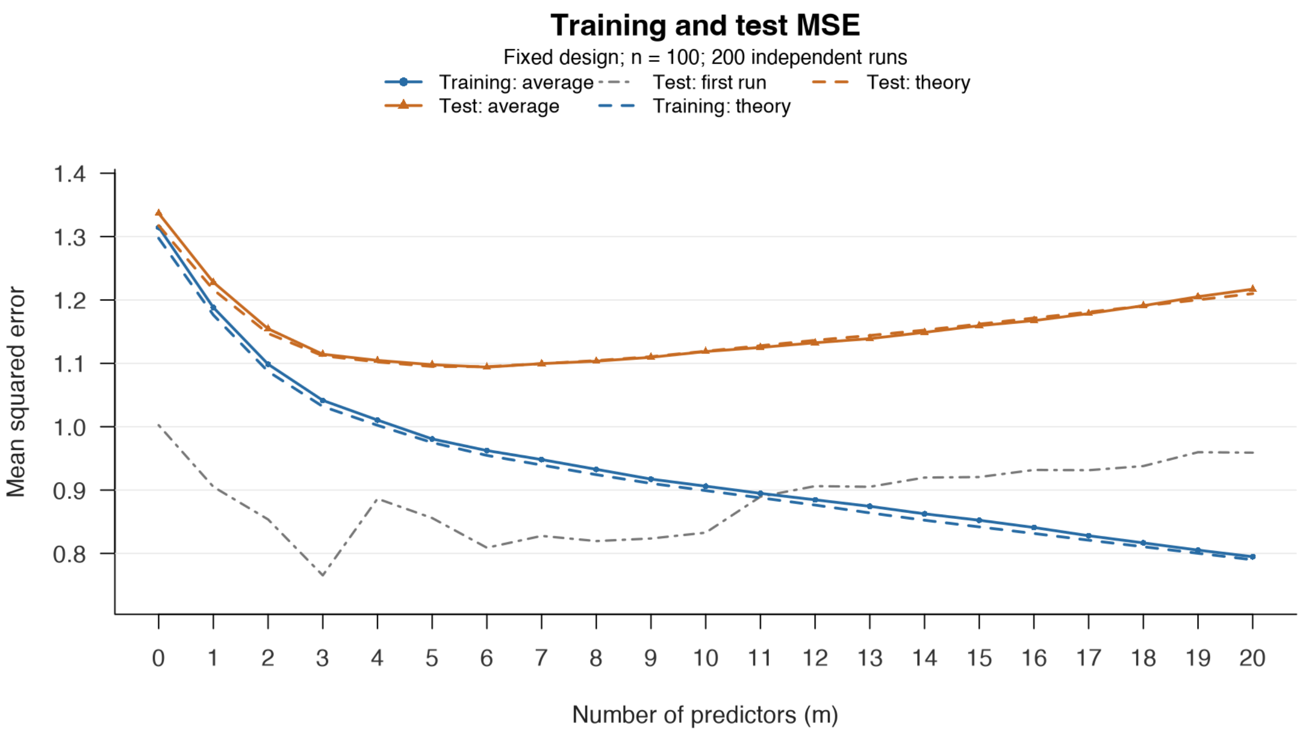

Figure 1. Training and test MSE for $m = 0, ..., 20$. Solid curves are averages over 200 runs; dashed curves are conditional expectations. The gray curve uses the first run only. The full numerical table is on the next page.

The average training MSE fell from **1.314664** at $m = 0$ to **0.794815** at $m = 20$. Average test MSE was lowest at ***m* = 6**, where it was **1.094007**; it increased to 1.217043 at $m = 20$.

I also checked all 20 consecutive changes in each of the 200 training curves. There were **0 increases out of 4,000 comparisons** (tolerance 10^{-10}). The largest change was about -4.13×10^{-11}. Adding a predictor expands the set of possible fits, so least squares cannot give a larger training RSS.

Question 1, continued

(b) Conditional expectations

For each model, I projected the fixed mean vector onto its design matrix to find the squared approximation bias. I used the realized design, not just the squared omitted coefficients.

$$B_m^2 = \|(I - H_m)\mu\|^2, \qquad H_m = X_m(X_m^\top X_m)^{-1}X_m^\top.$$

$$E(\mathrm{MSE}_{\mathrm{train},m} \mid X_{\mathrm{all}}) = \frac{B_m^2}{100} + 1 - \frac{m+1}{100}, \qquad E(\mathrm{MSE}_{\mathrm{test},m} \mid X_{\mathrm{all}}) = \frac{B_m^2}{100} + 1 + \frac{m+1}{100}.$$

m	Mean training	Expected training	Mean test	Expected test
0	1.314664	1.297425	1.336716	1.317425
1	1.188081	1.176263	1.227913	1.216263
2	1.098616	1.087190	1.154570	1.147190
3	1.041517	1.031905	1.114369	1.111905
4	1.010472	1.002207	1.104631	1.102207
5	0.980625	0.975007	1.097872	1.095007
6	0.962333	0.954702	1.094007	1.094702
7	0.948099	0.939493	1.099547	1.099493
8	0.932711	0.924306	1.103380	1.104306
9	0.917275	0.910531	1.109580	1.110531
10	0.906128	0.899164	1.118629	1.119164
11	0.894808	0.887772	1.125064	1.127772
12	0.884602	0.876365	1.132503	1.136365
13	0.874350	0.863990	1.138992	1.143990
14	0.862494	0.852436	1.148920	1.152436
15	0.852371	0.841897	1.159433	1.161897
16	0.840921	0.831490	1.167473	1.171490
17	0.827927	0.820908	1.178563	1.180908
18	0.816639	0.810605	1.190993	1.190605
19	0.805218	0.800238	1.205204	1.200238
20	0.794815	0.790000	1.217043	1.210000

The simulated averages are close to the theoretical curves. Their largest absolute differences are about 0.0172 for training and 0.0193 for test MSE. The expected test MSE also reaches its minimum at $m = 6$ (1.094702). At $m = 20$, the approximation bias is zero, giving expected training and test MSE of 0.79 and 1.21.

(c) One run and model size

The first-run test curve is more uneven because it depends on one pair of noise vectors. Its minimum is at $m = 3$, not 6. Averaging 200 runs reduces these random fluctuations. As predictors are added, approximation bias decreases, but the estimation-variance term $(m + 1)/100$ increases. Test error can therefore rise even while training error keeps falling. Selecting by training error alone would keep favoring larger models.

Question 2. Prediction at one target point

(a) Theoretical squared error

The target is $x_0 = (0, 1, 0, 0, 1, 0)$. Using the given formula:

$$\mu_0 = 0.5 + 0.5 = 1, \qquad \text{Bias}_m^2 = \left(\sum_{j=m+1}^{6} x_{0j}\beta_j \right)^2, \qquad \text{Var}_m = \frac{1 + \sum_{j=1}^{m} x_{0j}^2}{100}.$$

m	Squared bias	Variance	Expected error	Simulated error
0	1.00	0.01	1.01	1.018919
1	1.00	0.01	1.01	1.018919
2	0.25	0.02	0.27	0.267117
3	0.25	0.02	0.27	0.267117
4	0.25	0.02	0.27	0.267117
5	0.00	0.03	0.03	0.033759
6	0.00	0.03	0.03	0.033759

Omitted predictors create a systematic gap between the average prediction and the target mean. Estimating the intercept and slopes adds random variation. Squared error combines these two parts. It changes only when predictor 2 or 5 enters, so the equal-error groups are m = 0-1, 2-4, and 5-6.

(b) Simulation

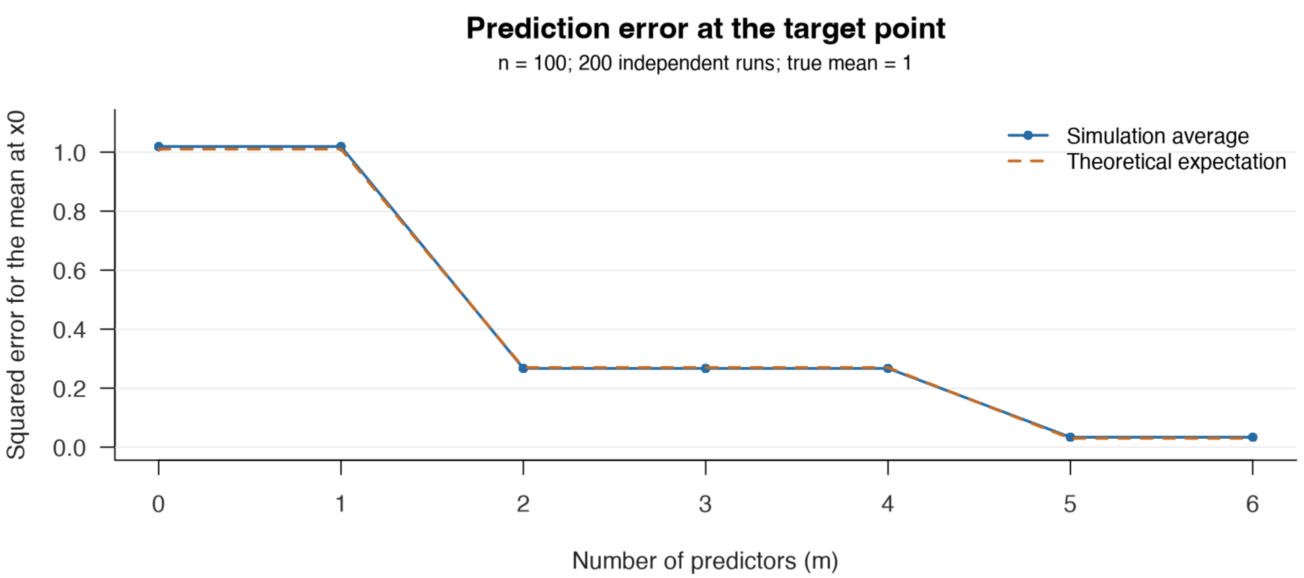

Figure 2. Squared error for the target **mean**, with 200 training-noise runs and seed 43203. The simulation agrees closely with the theoretical levels. The smallest theoretical minimizer is ***m* = 5**; m = 6 ties at 0.03.

(c) Why the target matters

Although every coefficient is nonzero, only coordinates 2 and 5 of the target are nonzero. The other four slopes are multiplied by zero when predicting this mean. With the orthogonal design, adding those predictors does not change the error here. Question 1 averages predictions over all 100 design rows and predicts noisy responses. This question concerns one mean, so there is no extra +1 term for new response noise.

Question 3. The optimism correction

(a) Derivation

All expectations below condition on the fixed design. The candidate model is chosen before observing the response. Write $k = p + 1$, including the intercept. The training residual can be written as:

$$y - Hy = b + (I - H)\epsilon, \qquad b = (I - H)\mu, \quad k = p + 1.$$

Zero-mean errors make the cross term vanish. Also, symmetry and idempotence give $(I - H)^T(I - H) = I - H$:

$$\begin{aligned} E\|y - Hy\|^2 &= B^2 + E[\epsilon^\mathsf{T}(I - H)\epsilon] \\ &= B^2 + \sigma^2 \operatorname{tr}(I - H) = B^2 + \sigma^2(n - k). \end{aligned}$$

For the test response, the two noise vectors are independent, so the cross terms again have expectation zero. Also, H is symmetric and idempotent:

$$\begin{aligned} y^* - Hy &= b + \epsilon^* - H\epsilon, \\ E\|y^* - Hy\|^2 &= B^2 + n\sigma^2 + \sigma^2 \operatorname{tr}(H) = B^2 + \sigma^2(n + k). \end{aligned}$$

Dividing both expressions by n gives the requested results:

$$E(\mathrm{MSE}_{\mathrm{train}} \mid X) = \frac{B^2}{n} + \sigma^2\left(1 - \frac{p+1}{n}\right), \qquad E(\mathrm{MSE}_{\mathrm{test}} \mid X) = \frac{B^2}{n} + \sigma^2\left(1 + \frac{p+1}{n}\right).$$

(b) Expected optimism

Subtracting cancels the approximation-bias and irreducible-noise terms:

$$E(\mathrm{MSE}_{\mathrm{test}} - \mathrm{MSE}_{\mathrm{train}} \mid X) = \frac{2(p+1)\sigma^2}{n} = \frac{10}{120} = 0.083333.$$

$$E(\mathrm{MSE}_{\mathrm{train}}) = 0.04 + 1 - \frac{5}{120} = 0.998333, \qquad E(\mathrm{MSE}_{\mathrm{test}}) = 0.04 + 1 + \frac{5}{120} = 1.081667.$$

This ordering concerns expectations, not every individual pair of responses. By chance, a test response can be closer to the fitted values than the training response is.

(c) Corrected MSE and Mallows' Cp

$$\widehat{\mathrm{MSE}}_{\mathrm{test}} = \frac{116.4 + 2(5)(0.96)}{120} = 1.05, \qquad C_p = \frac{116.4}{0.96} - 120 + 10 = 11.25.$$

The observed training MSE is 116.4/120 = 0.97, and the optimism adjustment is 0.08. Rearranging the definition of Cp gives:

$$\widehat{\mathrm{MSE}}_{\mathrm{test}} = \frac{\widehat{\sigma}^2}{n}(C_p + n) = \widehat{\sigma}^2 + \frac{\widehat{\sigma}^2}{n}C_p.$$

When n and the same positive noise-variance estimate are used for every candidate, this is an increasing linear transformation of Cp. Therefore, both criteria select the same minimizing model.

Question 4. Cp, AIC, and BIC

(a) Model fits

I used only rows 1-370 of diabetes.csv, with y as the response. All three models include an intercept. Model A uses bmi, bp, s5, sex, s1, s2, and s4; Model B adds s6. The reference model uses all ten predictors.

Model	Predictors (p)	RSS	Residual df
Model A	7	1098908.566	362
Model B	8	1089717.057	361
Full reference	10	1089451.721	359

The full model has 370 - (10 + 1) = **359** residual degrees of freedom. Its variance estimate is shared by Models A and B:

$$\widehat{\sigma}^2 = \frac{\mathrm{RSS}_{\text{full}}}{370 - 11} = \frac{1{,}089{,}451.721219}{359} = 3{,}034.684460.$$

(b) Model-selection criteria

I calculated the reduced criteria directly from the formulas, using natural logarithms:

$$C_p = \frac{\mathrm{RSS}}{\widehat{\sigma}^2} - n + 2(p+1), \qquad \mathrm{AIC}^{\text{red}} = n\log(\mathrm{RSS}/n) + 2(p+1),$$

$$\mathrm{BIC}^{\text{red}} = n\log(\mathrm{RSS}/n) + (p+1)\log n.$$

Model	Cp	Reduced AIC	Reduced BIC
Model A	8.116253	2974.640260	3005.948284
Model B	7.087434	2973.532485	3008.754012

Cp and reduced AIC select Model B. Reduced BIC selects Model A. These choices use comparisons within each column; the sizes of different criteria are not directly comparable.

(c) Fit improvement versus the extra penalty

Adding s6 reduces RSS by **9,191.508850**. For Cp, the reduction in scaled RSS exceeds the penalty of 2 for one extra fitted coefficient:

$$\frac{\mathrm{RSS}_A - \mathrm{RSS}_B}{\widehat{\sigma}^2} = 3.028819 > 2,$$

For the reduced AIC and BIC, the improvement in the shared log-RSS term is:

$$370\log\left(\frac{\mathrm{RSS}_A}{\mathrm{RSS}_B}\right) = 3.107775, \qquad 2 < 3.107775 < \log(370) = 5.913503.$$

That improvement exceeds the AIC penalty of 2 but falls short of the BIC penalty, log(370). So BIC favors the smaller model even though adding s6 improves the fit. Selection does not prove a model is the true data-generating model: the comparison uses a finite sample and a limited set of candidates, and the criteria balance fit and complexity differently.

Question 5. Validation and final test data

(a) Where the test data are misused

The first misuse is **Step 2**: the final test responses are opened to compare all eleven candidates before model selection is finished. Step 3 then uses those comparisons to choose a model. The test set is now serving as validation data, even though it was not used to estimate regression coefficients.

Each test MSE contains random error. Taking the minimum tends to pick a model that benefited from favorable noise on this particular test set. Reporting that same minimum as an independent final estimate is generally too optimistic.

(b) Five-fold cross-validation within the training set

Keep rows 371-442 (72 observations) untouched until the final evaluation. Use only the first 370 rows to choose among the eleven models, with predictors entering in the given order: age, sex, bmi, bp, s1, s2, s3, s4, s5, s6.

1. Randomly split the 370 training observations into five folds of 74. Use the same folds for every candidate model.

2. For each predictor count m = 0, ..., 10, fit the model on four folds (296 observations) and predict the remaining fold. Repeat until each fold has been held out once.

3. Average the 370 held-out squared errors for each m. Any data-dependent preprocessing must also be fitted using only the four training folds, then applied to the held-out fold.

$$\mathrm{CV}(m) = \frac{1}{370} \sum_{i=1}^{370} \left(y_i - \widehat{y}_{i,m}^{(-\mathrm{fold}(i))} \right)^2 .$$

4. Choose the predictor count with the smallest CV error. Decide a tie rule beforehand, such as choosing the smaller model. Refit that selected model on all 370 training observations.

5. Evaluate the refitted model once on the 72 final test observations and report its test MSE. Do not use that result to change the predictor count or tune the procedure again.

(c) If all eleven test errors have already been examined

The eleven errors and the chosen model can still be reported as exploratory results, with a clear explanation that this test set was used in model selection. The minimum should not be described as an untouched final evaluation. Running CV afterward does not make the old test set independent again.

For a new final evaluation, I would need fresh independent observations that have not been used to choose models or analysis settings. The model and evaluation procedure should be fixed before using that new data.

Computation and sources

The companion homework-02.R reproduces the calculations, tables, and plots in this document. It uses R, the stated seeds, 200 runs in Questions 1 and 2, and only the first 370 diabetes rows for Question 4. Values in the tables are rounded for display. Question 5 describes the validation procedure; it does not inspect or score the final test set.

Assignment and data: STAT 432, Week 02 homework. Reference: James, Witten, Hastie, Tibshirani, and Taylor, *An Introduction to Statistical Learning*, Chapters 3, 5, and 6.